MARCH 23, 2026

PIGGY PILOT SAVINGS

BUDGETTING APP

ST10455770 KAGISO MAPUTLA

POE PART1

Table of content:

1. Introduction

1.1 Overview of the App

1.2 App Name and Icon

1.3 Innovative Features

2. Requirements Specification

2.1 Functional Requirements

2.2 Non-Functional Requirements

3. User Interface Design

3.1 Screen Mockups

3.2 Purpose of Each Screen

3.3 Navigation Flow Diagram

4. Project Plan

4.1 Development Tasks

4.2 Milestones

4.3 Gantt chart

5. Conclusion

6. References

1. Introduction

Managing personal finances is a challenge for many people, especially young adults and students who may be new to budgeting. To address this issue, the **Piggy Pilot Saving Tracker** app is designed as an easy-to-use mobile application that helps users track income, expenses, savings, and financial goals. The app combines budgeting tools with gamification elements to encourage consistent saving habits.

The app's initial icon will feature a simple pink piggy-bank silhouette with coins dropping into it, symbolizing saving and financial growth. The design is friendly, minimalistic, and appropriate for all age groups.

Unlike traditional budget apps that only record transactions, **Piggy Pilot** introduces innovative features such as weekly challenges, reward badges, progress animations, financial tips, and spending-pattern insights. The purpose of this design document is to outline the full requirements, features, user interface, and project plan needed to develop the first working prototype of the app.

---

2. Detailed Requirements

2.1 <u>Functional Requirements</u>

1. **User Registration and Login**
    - Users must be able to create an account using email, username, and PIN.
    - Login should validate credentials and allow password/PIN reset.
    - User profile stored securely in a database.
2. **Add Income**
    - Users can capture weekly, monthly, or once-off income amounts.
    - Income types include salary, allowance, freelance earnings, etc.
    - App must automatically calculate total income for the period.
3. **Add and Track Expenses**
    - Users can record expenses under categories (food, transport, rent, entertainment, etc.)
    - Each expense must have: date, amount, category, optional notes.
    - App must calculate total expenses and remaining balance.
4. **View Spending Overview**
    - A dashboard screen must display:
        - Total income
        - Total expenses
        - Remaining balance
        - Top spending categories
    - Use simple graphs or bars (optional, non-3D to meet prototype scope).
5. **Savings Goals**
    - Users can create a savings goal with a target amount and deadline.

- App should track progress and show percentage completed.
- Users can add money towards goals.

6. **Gamification Features (Innovative Feature)**
    - Weekly savings challenges (e.g., “Save R50 this week”).
    - Reward badges for completing goals.
    - Streak system for logging activity daily.
    - Motivational notifications.
7. **Notifications**
    - App must send reminders to:
        - Log expenses
        - Stay below weekly limits
        - Continue savings streak
8. **Settings**
    - Users can update profile, currency settings (default ZAR), and notification preferences.
9. **Data Backup (Optional)**
    - Automatic backup to cloud or local storage so users do not lose progress.

2.2 Non-Functional Requirements

1. **Usability**
    - Simple, clean interface with easy navigation.
    - Large buttons, readable text, beginner-friendly design.
2. **Performance**
    - App should load within 2–3 seconds.
    - Dashboard calculations must update instantly.
3. **Security**
    - PIN-based login.
    - Secure storage of user data using encryption where applicable.
4. **Compatibility**
    - Must run on Android devices (Android Studio + Kotlin).
    - Works on small and large screens.
5. **Reliability**
    - App should not crash when adding transactions.
    - Validation for all inputs (no negative amounts, empty fields, etc.).

---

3. User Interface Design

3.1 Screen Mockups + Descriptions

Welcome / Splash Screen

Purpose:

- Introduces the app with the icon and name "Piggy Pilot"
- Loads the application

---

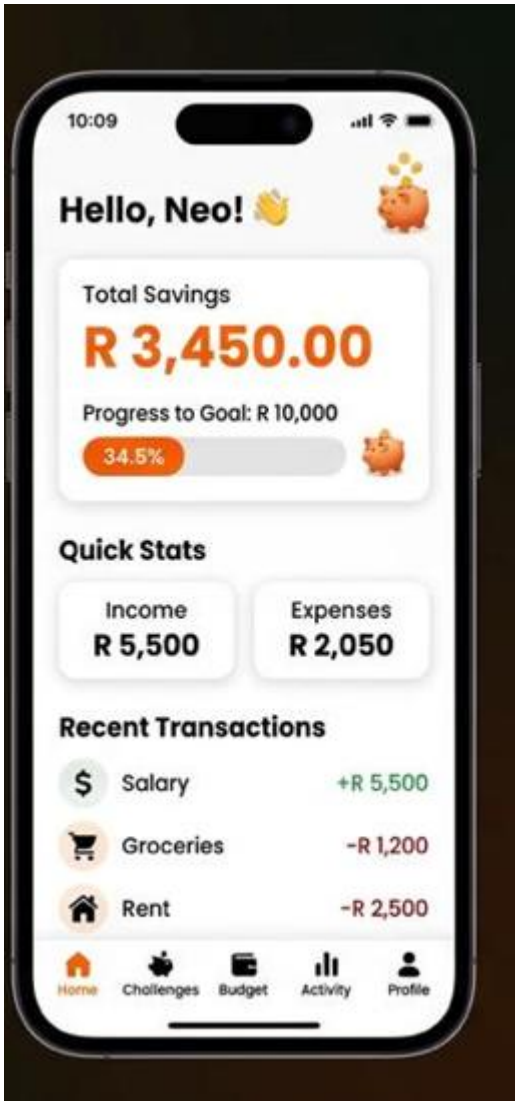
2. Login / Registration Screen

Elements: Email, Username, PIN, Login button, Register button
Purpose:

- Allows the user to sign up or log in.

---

3. Home Dashboard



Elements:

- Total income
- Total expenses
- Remaining budget
- Quick-add buttons (Add Income, Add Expense)
- Simple chart showing spending

Purpose:

- Gives the user a summary of their financial status.

---

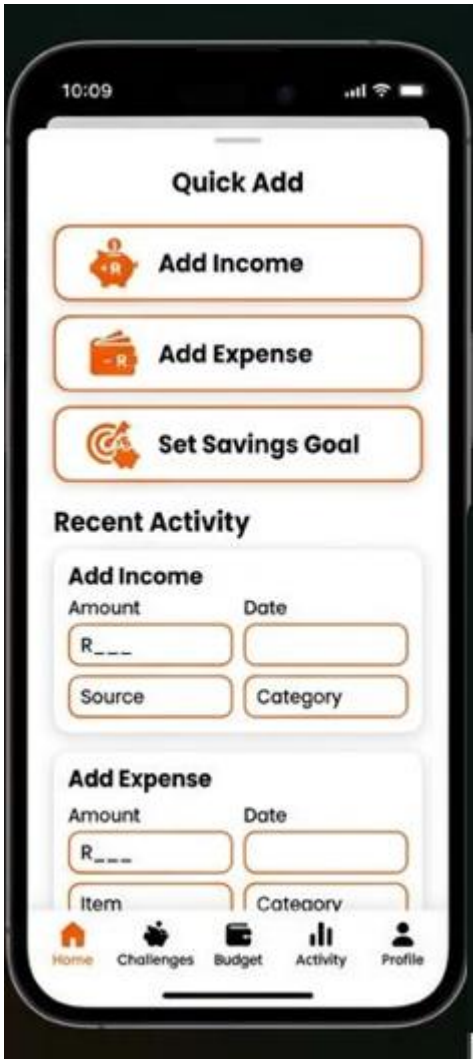
4. Add Income Screen

Purpose:

- Capture new income entries
- Calculate updated totals

---

5. Add Expense Screen

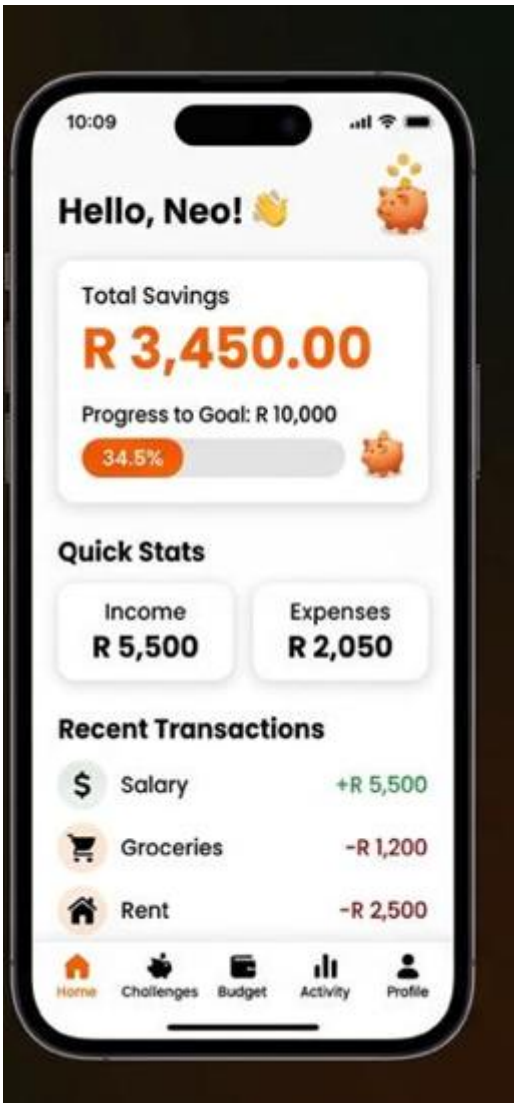


Purpose:

- Capture expenses with category selection
- Automatically update remaining balance

---

6. Savings Goals Screen

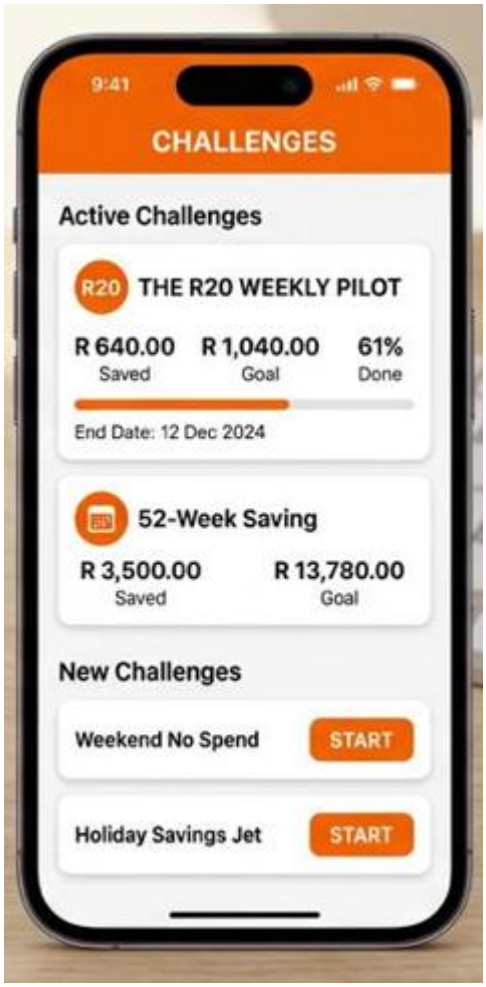


Purpose:

- Show list of goals
- Add goal or update progress
- Display percentage completed

---

7. Challenges & Badges Screen

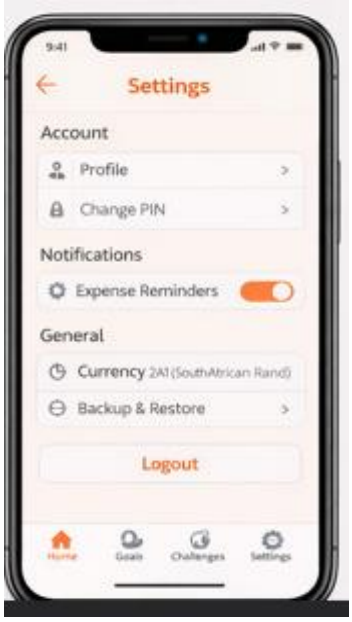


Purpose:

- Display weekly challenges
- Unlockable badges
- Track streaks

---

8. Settings Screen



Purpose:

- Update user info
- Change notification preferences
- Switch currency (optional)

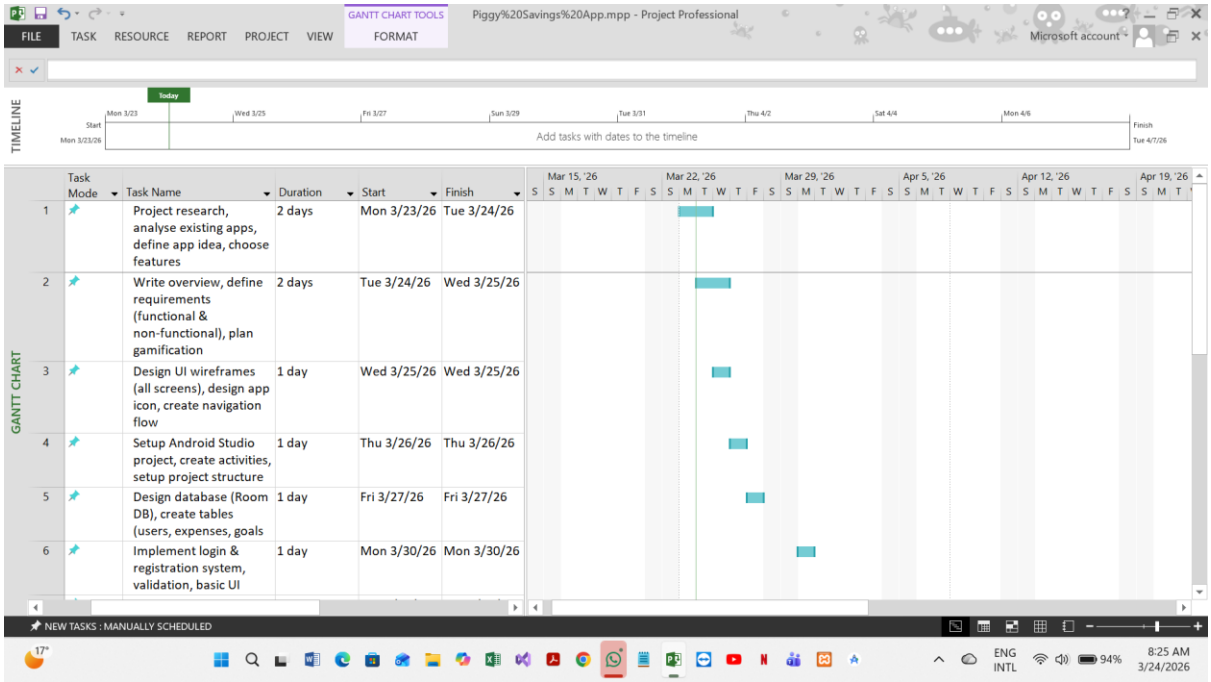
3.2 Navigation Flow Diagram

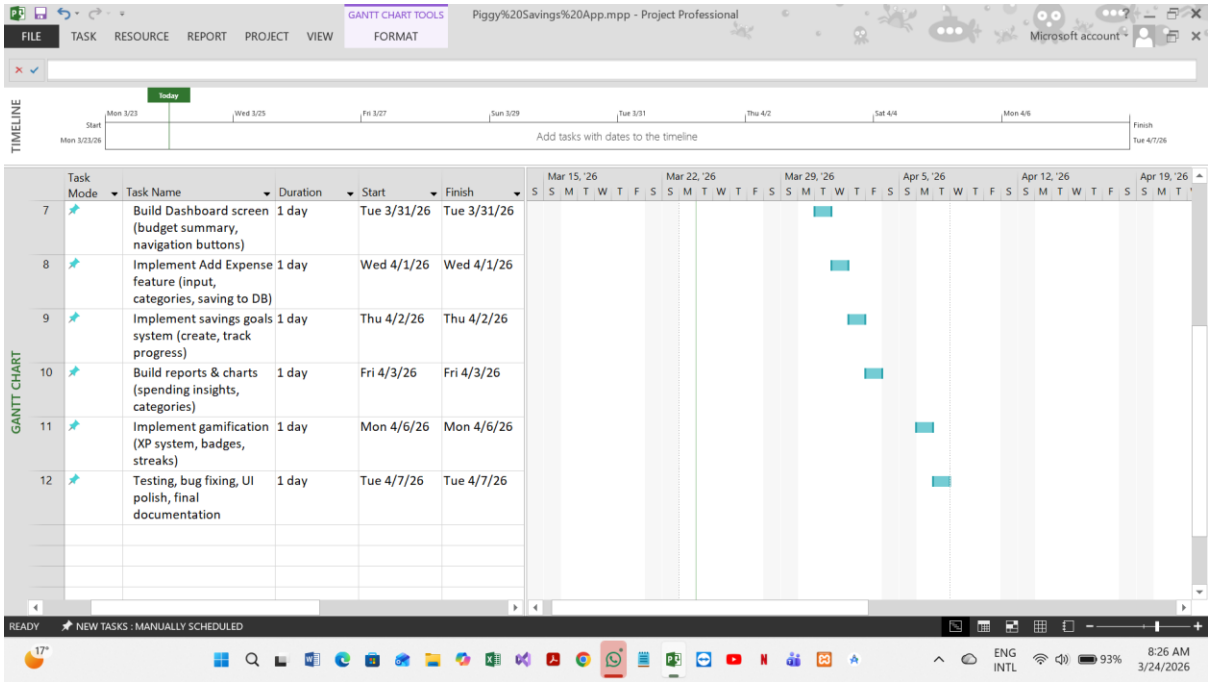
The user can navigate back to the dashboard from all screens using a back button or the bottom navigation bar.

---

4. Project Plan – Gantt chart

Below is a breakdown of tasks for a **12-day** development plan:





5. Conclusion

The Piggy Pilot Budget Tracker app is designed to be a simple yet engaging financial management tool. Through its combination of budgeting functions and gamification, the app helps users develop good financial habits and stay motivated. This document outlines the requirements, design, and development plan needed to create a functional prototype. By following these detailed specifications, developers will be able to build the app efficiently and accurately.

6. References

- Goodbudget App Review (2024)
- Vault22 Features & User Ratings (2024)
- Expenses OK Official App Page
- UX Design Principles for Mobile Apps (2023)
- Expenses OK. (2026). Expenses OK on Google Play. Available at: https://play.google.com
- Goodbudget. (2026). Goodbudget Budget Planner. Available at: https://goodbudget.com
- Vault22. (2026). Vault22 on Google Play.